

WASEDA University
Graduate School of Information, Production and Systems

Diagnosis-Guided Augmentation and Frequency-Gated Fusion for Actor-Conditioned Human-Value Recognition

Peiyang Huang

Midterm Presentation · Focus: WAIPS-03

Data Engineering Lab, IPS, Waseda University

Whose values does a news report describe?

Chile protests: a reported government statement

> *"He justified the decision, saying he wanted to maintain **public order** against vandals and crime."*

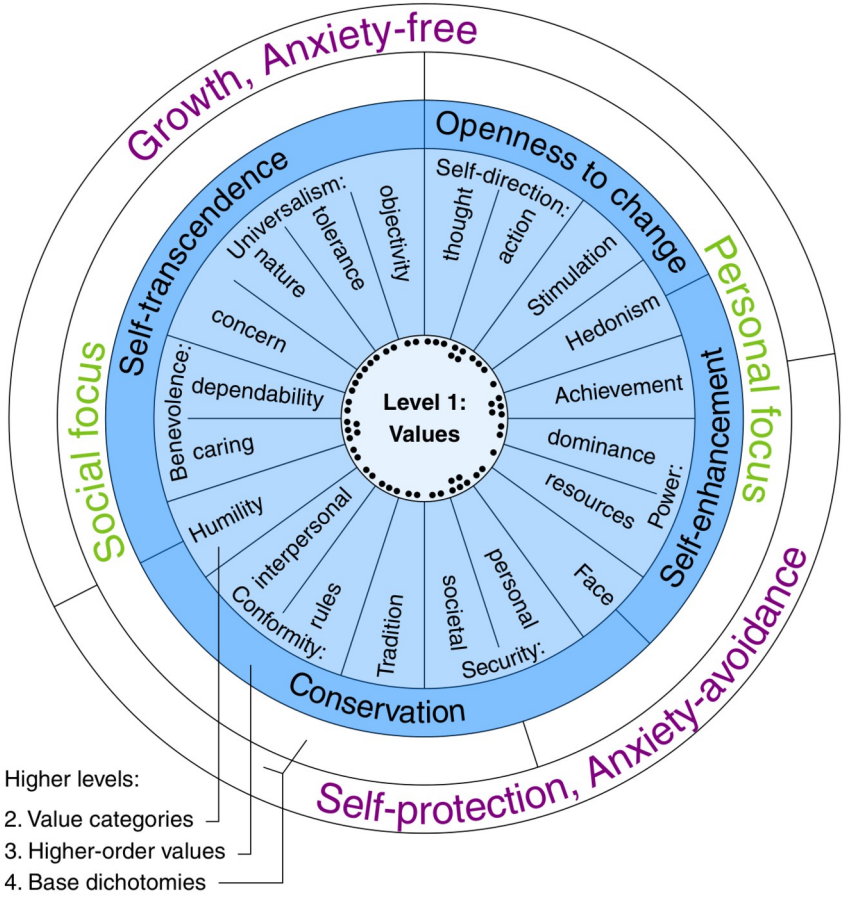
Whose conduct?	Which value?	Which direction?
Government and Security Forces	Have a safe country	**Aligned**

Aligned = supports the value. Contradictory = goes against it.

The benchmark labels reported conduct in its news context.

Source excerpt and reference: Huang, section 5.9, Table 9 (Chile case).

A shared vocabulary for human values



54 specific values

20 broader categories

Values describe what people consider important.

Example: Security, societal

Have a safe country

Have a stable society

Kiesel et al. (ACL 2022),
mainly based on Schwartz's value theory.

Figure: Kiesel et al., ACL 2022, Fig. 1. aclanthology.org/2022.acl-long.306/

NTCIR-19 FEHU: a shared evaluation task

Fact-based Event-centric Human-value Understanding

Values in factual news, at article and event levels

Organizing institutions

University of Tsukuba
Yao Wang, Zhuochen Liu,
Jiankang Chen, Haitao Yu

AIST (Japan)
Xin Liu, Kyoungsook Kim

University of Innsbruck
Adam Jatowt

Shared-task timeline

Date	Milestone
Sep 2025	Task launch
Jan 2026	Training data
Aug 2026	Results returned
8–10 Dec 2026	Conference, Tokyo

Sources: FEHU organizers / important dates; NTCIR-19 conference. Checked 1 Oct 2026.

My task: labels for each supplied actor

Input

News article

Supplied actor list



For each actor

Relevant value + direction

Aligned / contradictory

Several labels may apply

54 values × 2 directions = 108 possible labels

Evidence comes from the article; actor identities are supplied.

Task definition: FEHU official resources; Huang, section 3.1.

What these labels could help readers do

Find reports about a value and an actor

"Find reports about public security and government actions."

Compare how reports portray institutions

Keep each judgment connected to the right person or organization.

A useful label must preserve who did what.

Application motivation: Huang, introduction. Reader question is illustrative.

Research goal: fewer wrong and missed labels

Miss a relevant value

The model may overlook
a value in an actor's conduct.

Add an incorrect label

A plausible interpretation
may disagree with the reference.

Questions for this talk

How should support from different predictions guide revision?

Which revision steps reduce errors?

Goal: recover relevant labels while controlling incorrect predictions.

Huang, sections 1 and 3.3. This talk focuses on the fusion research question.

How the classifier supports the decision

Approach	Role of the trained classifier
EAVIT	Narrow value definitions before the language model decides
WAIPS-03	**Help revise labels after the initial predictions**

Label selection and direction checking use separate decisions.

My design uses classifier scores to support selective revision.

Zhu et al., EAVIT (IJCAI 2025); Huang, section 2. doi.org/10.24963/ijcai.2025/934

Data augmentation for missed values

Step	Purpose
Diagnose errors	Find 18 directed labels missed by the initial classifier
Generate examples	Connect an actor's conduct to a target value and direction
Train DeBERTa	Learn from official data and generated examples

In fusion, an earlier augmented classifier supplies candidate scores.

Generated examples help, but standalone recognition remains difficult.

Huang, sections 4.2–4.3 and Table 5. Independent repeated training results: backup 22.

WAIPS-02: prompt agreement and direction choice

Step	Decision
Three prompts	Analyze the same article with different demonstrations and instructions
Majority vote	Keep a directed label when at least two prompts agree
Direction choice	Use an independent prediction to choose the direction

A correct label found by only one prompt can be lost.

This separates value inclusion from direction choice.

Huang, section 4.4: three Pro prompt variants, followed by a Flash direction lookup.

WAIPS-03 combines scores and prompt predictions

PROPOSED METHOD · ACTUAL INPUTS AND THEIR ROLES

Input	Role in fusion
Initial prediction	Propose labels from generated event descriptions
Augmented classifier	Score candidates using learned label patterns
Four auxiliary predictions	Supply candidates and agreement, including Liu's targeted prompt

WAIPS-02 informs the design: select values first, then check direction.

Its final output is not an input to WAIPS-03.

Fusion decides which initial labels to revise using these supporting signals.

Huang, sections 1.3 and 4.5–4.7. WAIPS-03 = frequency-gated fusion, E7-G4.

Selective revision limits both types of label error

Operation	Reason	Constraint
Remove	Reduce weakly supported predictions	Keep at least one initial label
Add	Recover supported missing candidates	At most 2 new labels per actor

Only labels with >130 training positives can be removed or added.

Rare labels keep their initial presence or absence.

Support guides revision, with a frequency rule for rare labels.

Huang, Algorithm 1 and sections 4.5–4.6. Exact formulas and settings: backup 25.

Direction checks triggered by disagreement

METHOD · A FOCUSED QUESTION AFTER LABEL SELECTION

Trigger

The current output has one direction for a value.

An auxiliary prediction supports the opposite direction.

> *"Does this actor's reported conduct support or contradict this value?"*

Input: article + actor + value. The current direction is withheld.

Skip values already assigned both directions.

Recheck a disputed direction while preserving the selected values.

Huang, section 4.6. Direction-only QA; the question shown summarizes the implemented prompt.

How I judge whether a method helps

EVALUATION SETUP · SPLITS AND MEANING OF THE SCORES

Training	Development	Official test
1,228 articles	177 articles	175 articles
Learn	Compare designs	Evaluate submissions

Micro-F1 Overall balance of wrong predictions and missed labels

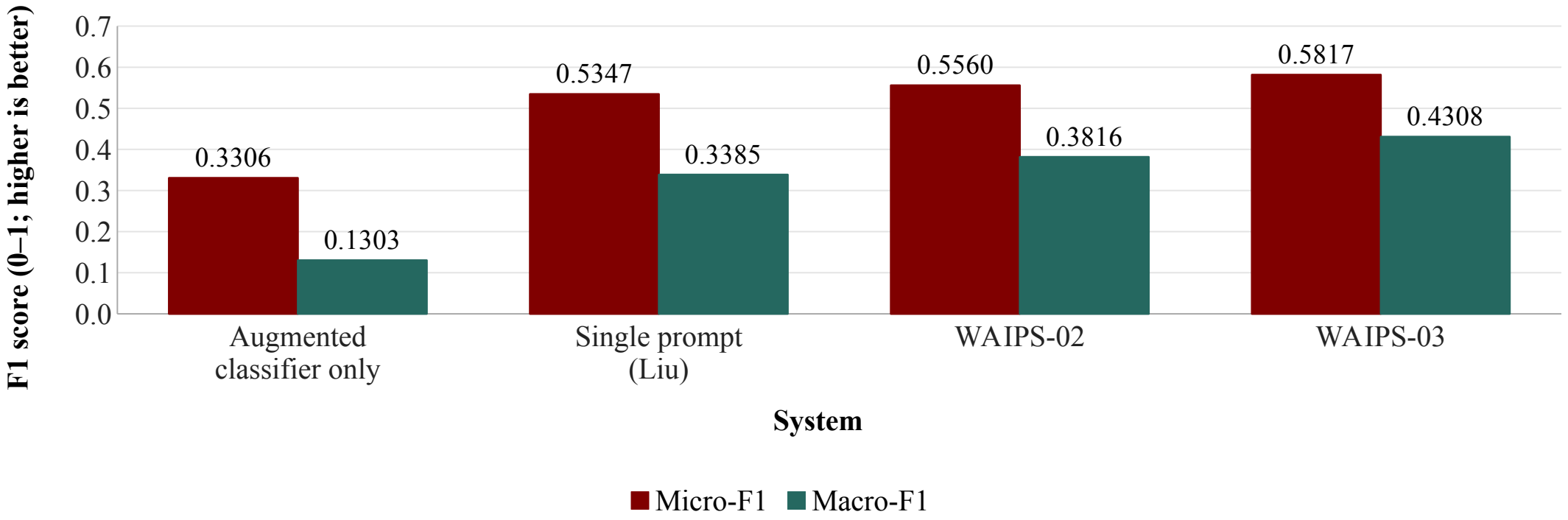
Macro-F1 Equal weight for each label with reference examples

A match requires the correct actor, value and direction. Scores: 0–1.

Huang, sections 5.1–5.3, Tables 2 and 5. FEHU official evaluator.

Four approaches on development news

EVALUATION · TASK 1b, 177 ARTICLES



Historical encoder: 456 labeled actors. Other systems: 459 supplied actors.

The 3 additional actors have no positive gold labels. Model-call budgets differ.

WAIPS-03 has the highest F1 among these four retained development results.

Huang, Table 4; archived Liu_frozen metrics (E7 report §1). Single retained outputs.

Fewer extra labels and fewer missed labels

Label counts	Initial	Final	Change
Predictions unmatched to reference	1,442	1,272	−170
Reference labels missed	938	888	−50

WAIPS-03 before and after revision

Same 177 development articles, matching actor + value + direction

Both error counts decrease, although many errors remain.

Derived from saved G0/G4 predictions against official development labels; consistent with Huang, Table 7.

Official test: better F1, remaining direction errors

EVALUATION · SEPARATE TEST SET, 175 ARTICLES

Submitted system	Micro-F1 ↑	Macro-F1 ↑
WAIPS-02: majority and direction choice	0.5496	0.3580
WAIPS-03: proposed fusion	**0.5808**	**0.3952**

Direction reversal rate rises: 0.0395 to 0.0477.

More predictions oppose the reference direction.

Overall recognition improves, while direction reliability remains a priority.

Official FEHU overview, Table 6; Huang, Table 8. Scores 0–1; lower reversal rate is better.

A direction check can introduce an error

Actor: Government and Security Forces · Value: Have a safe country

> *"...he wanted to maintain **public order** against vandals and crime."*

Before checking	After checking	Reference
Aligned	Contradictory	Aligned

The saved answer contains the direction, without an explanation.

Direction revision can overturn a correct initial judgment.

Huang, section 5.9, Table 9: Chile, 0-4716-3-1, actor -1, value 26. Excerpt shortened.

Next: evidence for more reliable direction changes

FUTURE WORK · PLANNED VALIDATION

A proposed change should cite the actor's conduct

Compare useful corrections and introduced errors
with their supporting passages.

Test on separate evaluation articles

Measure harmful revisions and missed corrections,
alongside overall label recognition.

The aim is fewer harmful changes while keeping useful corrections.

Planned work, extending Huang, sections 6.1–6.2. Effectiveness is not yet established.

Beyond FEHU: value analysis across news sources

FUTURE WORK · PLANNED TRANSFER AND APPLICATION

Research step	Question to test
New sources and events	Does revision still help on news outside the FEHU collection?
Researcher review	Do labels and supporting passages help compare actors' conduct?
Human evaluation	Are labels and passages correct? Does review take less time?

Test transfer to new material and usefulness for human analysis.

Proposed research extension based on Huang, §6.2 evidence evaluation; no transfer result yet.

Conclusion: selective fusion improves recognition

Research process

Training augmentation and prompt aggregation inform WAIPS-03.
Fusion combines classifier scores with auxiliary predictions.

Main result

Best development F1 among the four approaches shown
and higher official-test F1 than WAIPS-02

Next: stronger text evidence and evaluation beyond FEHU.

Huang, Tables 4 and 7–9; archived Liu_frozen results. Main talk ends here.